



ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 10

TECHNICAL SCIENCES

PAPER 1 (PHYSICS)

MARKS: 100

TIME: 2 hours

Topics: units and measurement, motion, energy and power, waves and sound, electric circuits, magnetism.

Written by Academic Central for practice, following the CAPS curriculum for Grade 10. It is not a Department of Basic Education examination paper.

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions. Show formulae, substitutions and units in ALL calculations. Use $g = 9,8 \text{ m}\cdot\text{s}^{-2}$.

$v_f = v_i + a\Delta t$	$\Delta x = \frac{v_i + v_f}{2} \Delta t$	$W = F\Delta x$	$P = \frac{W}{\Delta t}$
$E_p = mgh$	$v = f\lambda$	$R = \frac{V}{I} \quad P = I^2 R$	$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2}$

QUESTION 1: MULTIPLE CHOICE

- 1.1 The SI unit of force is the ...
 A joule. B newton.
 C watt. D pascal.
- 1.2 Which instrument measures the diameter of a thin wire most accurately?
 A Metre rule B Micrometer screw gauge
 C Measuring tape D Ammeter
- 1.3 The rate at which work is done is called ...
 A energy. B force.
 C power. D momentum.
- 1.4 An ammeter must be connected in a circuit ...
 A in parallel. B in series.
 C either way. D across the battery only.
- 1.5 Sound travels FASTEST through ...
 A air. B water.
 C steel. D a vacuum.

(5 × 2)

[10]

QUESTION 2: UNITS AND MEASUREMENT

- 2.1 Write down the SI unit of: (a) mass; (b) time; (c) electric current. (3)
- 2.2 Convert: (a) 2,5 kN to N; (b) 450 mm to m; (c) $72 \text{ km}\cdot\text{h}^{-1}$ to $\text{m}\cdot\text{s}^{-1}$. (4)
- 2.3 Write 0,000 32 m in scientific notation. (2)
- 2.4 On a micrometer screw gauge, the main scale shows 5,5 mm and the thimble scale shows 28 divisions. Each thimble division is 0,01 mm. Calculate the reading. (3)

[12]

QUESTION 3: MOTION

- 3.1** A delivery truck drives 12 km north and then 5 km south. The trip takes 20 minutes.
- 3.1.1 Write down the total distance travelled. (1)
 - 3.1.2 Write down the truck's displacement. (2)
 - 3.1.3 Calculate the average speed in $\text{km}\cdot\text{h}^{-1}$. (3)
 - 3.1.4 Calculate the average velocity in $\text{km}\cdot\text{h}^{-1}$. (3)
- 3.2** A car accelerates uniformly from $10 \text{ m}\cdot\text{s}^{-1}$ to $30 \text{ m}\cdot\text{s}^{-1}$ in 5 s.
- 3.2.1 Define *acceleration*. (2)
 - 3.2.2 Calculate the acceleration of the car. (3)
 - 3.2.3 Calculate the distance covered in the 5 s. (3)
- 3.3** Explain the difference between a scalar and a vector, with ONE example of each. (3)
- [20]**

QUESTION 4: ENERGY AND POWER

An electric hoist in a workshop lifts a 200 kg engine vertically through 3 m at a constant speed in 12 s.

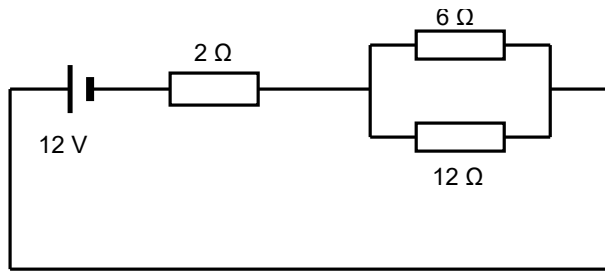
- 4.1 Calculate the weight of the engine. (2)
 - 4.2 Calculate the work done by the hoist on the engine. (3)
 - 4.3 Define *power* and calculate the useful power of the hoist. (5)
 - 4.4 Write down the gravitational potential energy gained by the engine. (2)
 - 4.5 The hoist is only 70% efficient. Calculate the electrical power it uses. (3)
 - 4.6 State the principle of conservation of energy, and name the main energy change in the hoist's motor. (3)
- [18]**

QUESTION 5: WAVES AND SOUND

- 5.1 Explain the difference between a longitudinal and a transverse wave. Give ONE example of each. (4)
 - 5.2 A sound wave has a frequency of 200 Hz and a wavelength of 1,7 m. Calculate its speed. (3)
 - 5.3 Explain how ultrasound can be used to find cracks inside a metal part. (2)
 - 5.4 Explain why workers must wear hearing protection near loud machines. (3)
- [12]**

QUESTION 6: ELECTRIC CIRCUITS

A 12 V battery (no internal resistance) is connected to a 2Ω resistor in series with a parallel combination of a 6Ω and a 12Ω resistor.



- 6.1 Define *resistance*. (2)
- 6.2 Calculate the effective resistance of the parallel part. (3)
- 6.3 Write down the total resistance of the circuit. (1)
- 6.4 Calculate the current from the battery. (3)
- 6.5 Calculate the potential difference across the $2\ \Omega$ resistor. (2)
- 6.6 Calculate the current through the $12\ \Omega$ resistor. (3)
- 6.7 Calculate the power dissipated in the $2\ \Omega$ resistor. (3)
- 6.8 Explain how a fuse protects an electrical appliance. (3)

[20]

QUESTION 7: MAGNETISM

- 7.1 Draw the magnetic field pattern around a single bar magnet. Show the direction of the field lines. (3)
- 7.2 Give TWO ways to make an electromagnet stronger. (2)
- 7.3 Give ONE industrial use of an electromagnet. (1)
- 7.4 Name TWO ferromagnetic materials. (2)

[8]

TOTAL: 100

END OF QUESTION PAPER